

A Coupled 2D NLSWE Regional-Propagation and 3D URANS–VOF Coastal-Impact Framework for Reconstructing the 2011 Tōhoku Tsunami

Research snapshot from an ongoing framework-development programme. The work presented establishes and verifies the computational foundation for future historical validation, coastal-impact studies and tsunami-defence assessment; it does not claim a fully validated reconstruction of the 2011 event. Rhyen Jivan | Summer Studentship Project

Real-event reconstruction



Figure A. Regional context of the 2011 Tōhoku earthquake and Kamaishi Regional2D corridor. Base map adapted from Encyclopædia Britannica; corridor and Kamaishi annotations added for this study.

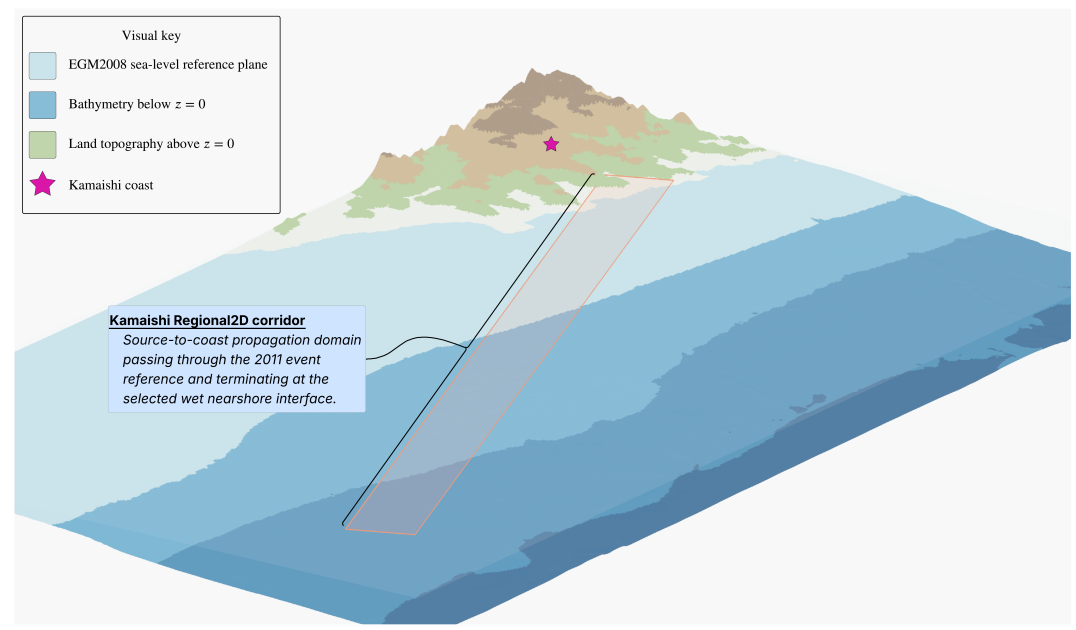
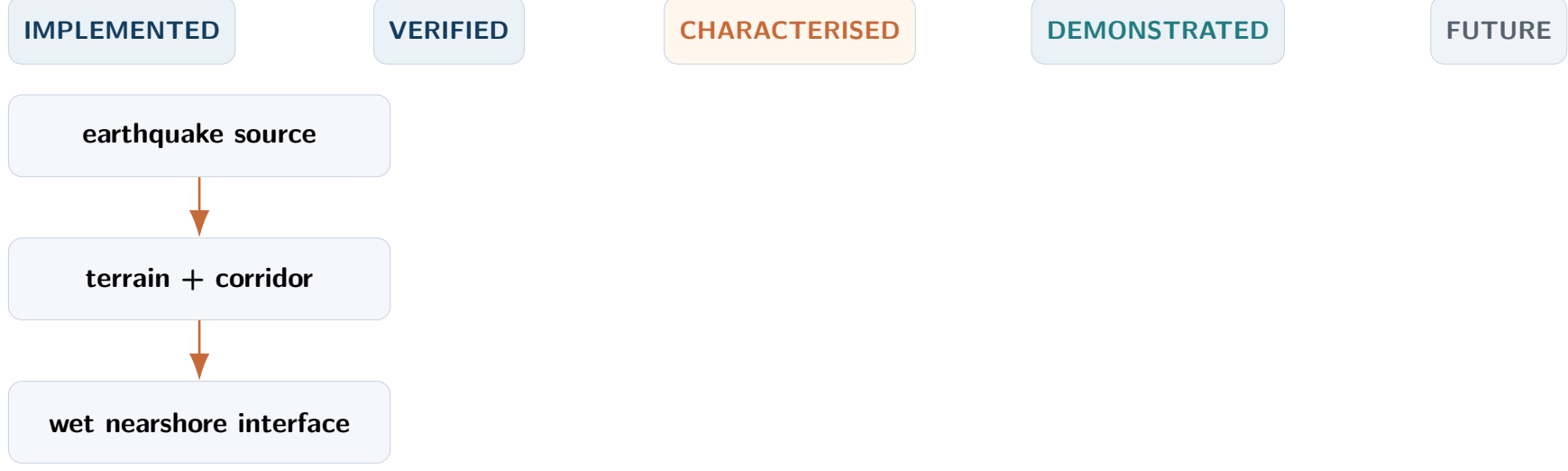


Figure B. Oblique bathymetry and topography of the Kamaishi Regional2D corridor. NOAA/NCEI ETOPO 2022, EGM2008; prepared in QGIS/GDAL and rendered in Blender 5.2 EEVEE with 1× vertical exaggeration.

Kamaishi target case

Kamaishi is used as the target coastal case study for reconstructing source-to-coast tsunami propagation from the 2011 Tōhoku event. The Regional2D corridor passes through the event reference and carries the wave toward the selected wet nearshore interface.



The real-event panel is organised as a visual sequence: regional context first, then oblique corridor geometry and the longitudinal bathymetric profile.

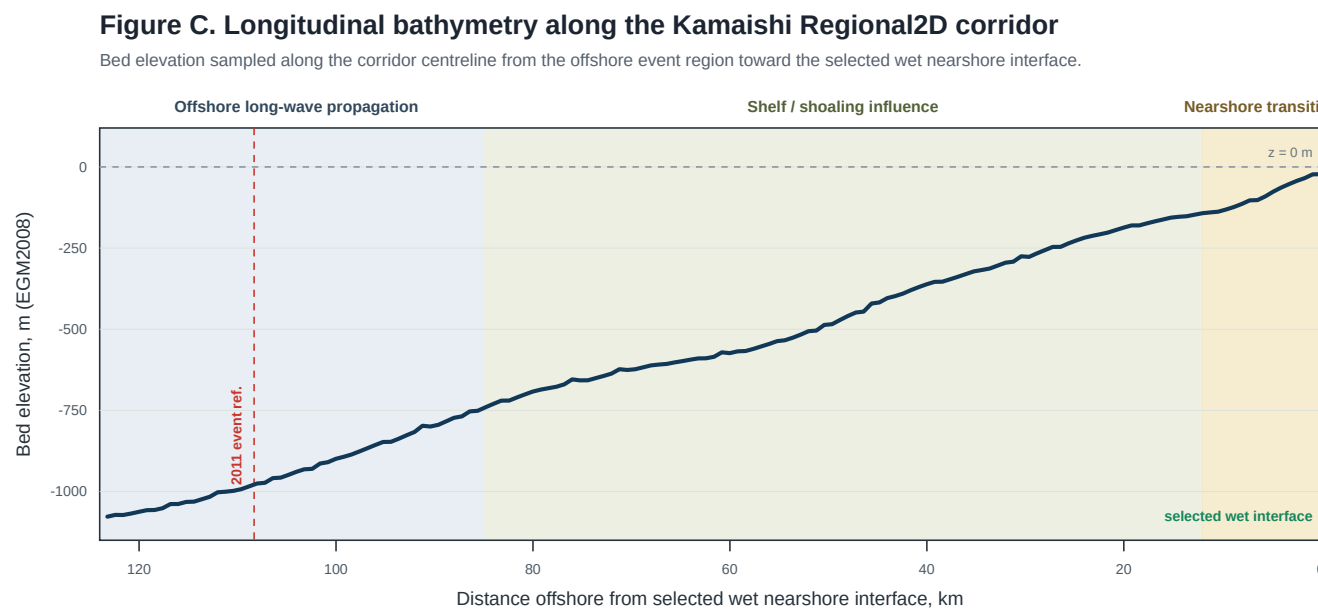


Figure C. Longitudinal bathymetry along the Kamaishi Regional2D corridor. Profile sampled along the accepted corridor centreline; bed elevation referenced to EGM2008.

Variables & quantities of interest

Solved / exported fields

Regional propagation state

- η free-surface elevation
- h water depth
- q_x, q_y depth-integrated discharge
- u, v depth-averaged velocity
- b bed elevation / bathymetry
- Δz_b coseismic bed displacement

Local flow / material state

- $\mathbf{U}$ 3D velocity
- p pressure
- α_{water} water volume fraction
- ρ, μ, μ_t mixture density and viscosities
- k, ω SST turbulence variables
- Re, y^+ diagnostic regime and wall resolution

Derived / future study metrics

Loading & interaction

- τ_w wall shear stress
- $F(t), M(t)$ resultant force and moment
- J force impulse
- pressure loading; viscous traction

Coastal / defence response

- run-up; inundation depth
- overtopping discharge / volume
- transmission; momentum transmission
- mechanical-energy transmission / dissipation

Response metrics define the later defence-study vocabulary; they are not all validated 2011-event results.

Validation framework

Observation targets

- offshore/deep-ocean waveform;
- domain-compatible Kamaishi/offshore observations;
- run-up;
- inundation.

VALIDATION FRAMEWORK ESTABLISHED 0 DIRECT 1 PROXY 28 TARGET ONLY

Primary validation uses no arbitrary time shift or amplitude scaling. Direct collocated waveform validation is not yet complete.

Metrics

- RMSE; MAE; Bias; correlation;
- peak-amplitude error;
- peak-time error.

Solution algorithms

Regional2D

- event + terrain state
- wet/dry classification
- limited-linear η, u, v reconstruction
- hydrostatic reconstruction
- Rusanov numerical face flux
- bed, Manning and boundary sources
- CFL-controlled timestep
- SSPRK3 update
- positivity, diagnostics and output

Pseudocode strip. For each regional step: reconstruct face states; apply hydrostatic treatment; compute fluxes and source terms; advance SSPRK3 stages; enforce admissibility; publish diagnostics. For each replay step: update mapped inlet forcing; advance OpenFOAM URANS–VOF state; monitor boundedness and Courant limits; extract free-surface and impact diagnostics.

The repository configures and audits OpenFOAM Foundation 11; it does not reimplement OpenFOAM pressure-correction, VOF or turbulence internals.

Local3D: adopted backend

- regional replay boundary data
- η, q_n, q_t inlet reconstruction
- $\alpha_{\text{water}} + \mathbf{U}$ boundary state
- OpenFOAM incompressible two-phase URANS–VOF
- pressure / velocity coupling
- VOF interface transport
- $k-\omega$ SST turbulence
- Courant and interface-Courant control
- fields, probes, $F(t), M(t)$

Why this matters

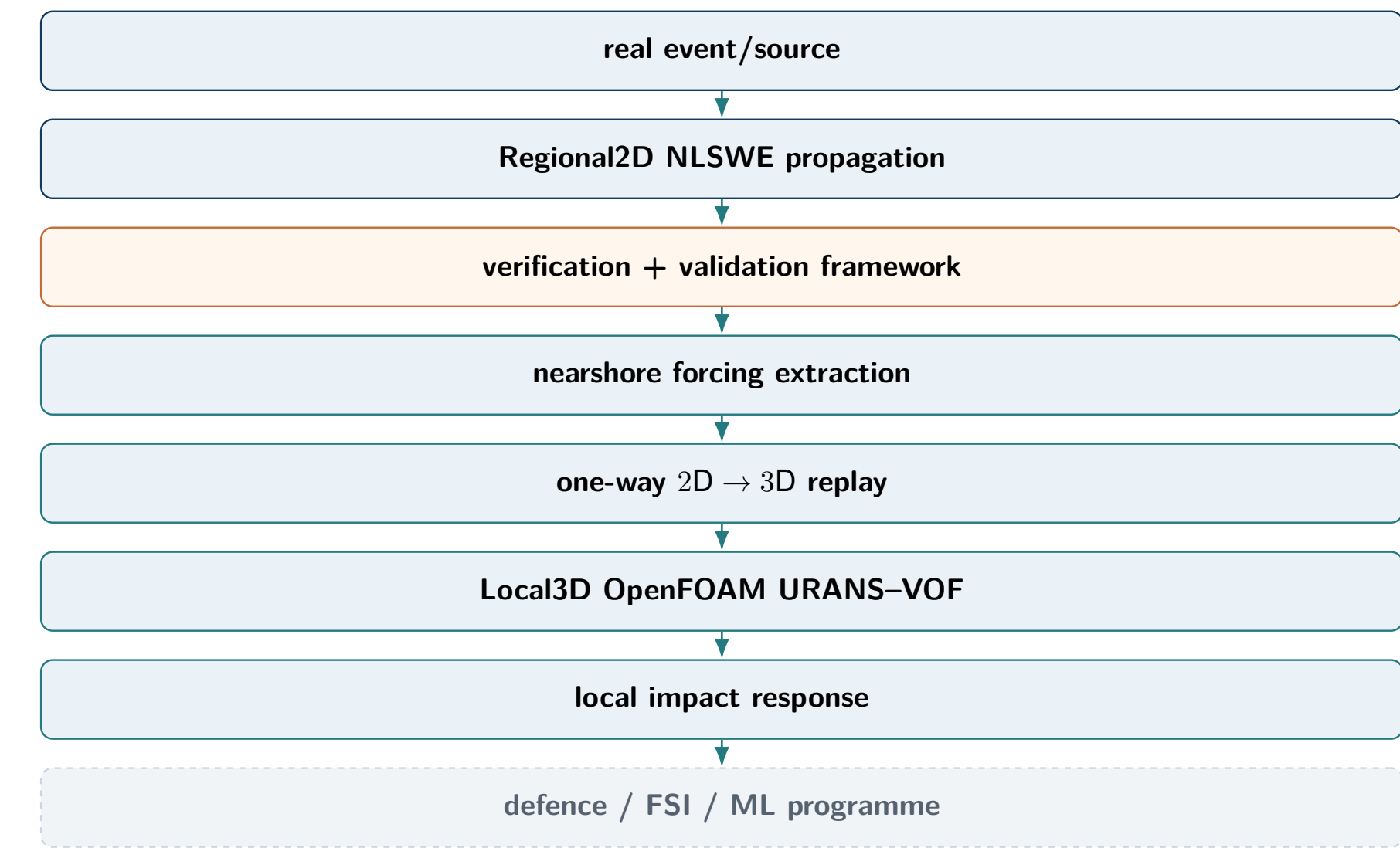
Defence assessment needs local loading and impact physics while preserving the link to the real tsunami source. Full source-to-coast 3D simulation is impractical, but purely depth-averaged modelling cannot resolve vertical impact, free-surface deformation or structure loading. Hybrid modelling bridges those scales.

Aim and objectives

Aim: Establish a computational framework for reconstructing and validating real tsunami events, then reuse validated event forcing for Local3D coastal-impact and tsunami-defence studies.

- Reconstruct the 2011 Tōhoku source using real event and terrain data.
- Propagate the event through a verified Regional2D NLSWE corridor.
- Establish observation-based historical-validation methodology.
- Demonstrate one-way transfer into a Local3D URANS–VOF impact model.
- Define response quantities for defence, resilience and optimisation.

Hybrid framework overview



System architecture only: implemented and verified modelling stages are separated from planned defence, FSI and data-driven extensions.

Verification method

Model / implementation consistency: research formulation → implementation traceability → zero required mismatches.

Controlled Regional verification:

- manufactured solution → FIRST ORDER BASELINE VERIFIED
- limited-linear reconstruction → SECOND ORDER LIMITED-LINEAR VERIFIED
- smooth-wave benchmark: amplitude error reduced 5.35×; phase-proxy error reduced 17.43×.

Real-event numerical fidelity:

- real-event refinement and projection-only diagnosis completed;
- terrain/source fidelity is the characterised dominant limitation;
- current real-event forcing is **BEST AVAILABLE – NUMERICALLY UNCERTAIN**.

Hybrid / Local3D evidence: forcing/replay consistency, boundary-reflection benchmark, VOF/runtime boundedness evidence, y^+ wall-treatment checks, probes and force/moment extraction.

Current achievements

- Regional2D implementation traced consistently to the research formulation.
- First-order baseline and limited-linear second-order formulation formally verified.
- Real 2011 Tōhoku → Kamaishi Regional2D reconstruction completed.
- Terrain/source fidelity characterised as the present real-event spatial limitation.
- One-way Regional2D→Local3D replay framework implemented and demonstrated.
- Accepted no-defence and simple-rigid-barrier Local3D replay evidence includes free-surface/probe output and force/moment extraction.
- Historical observation catalogue and validation methodology established.

Next steps

- Improve event/source and terrain fidelity, and refine or extend the validation domain where observation collocation requires it.
- Perform uncalibrated historical comparison, followed by documented calibration and holdout validation.
- Resolve current-generation Local3D VOF replay boundedness before adopting latest regional forcing for impact studies.
- Execute systematic no-defence / defence impact comparisons using agreed loading, overtopping, inundation and transmission metrics.
- Extend toward FSI, structural resilience, impact-response databases and future ML-based optimisation/design.

References

Encyclopædia Britannica; Japan Earthquake and Tsunami of 2011; Encyclopædia Britannica; n.d. NOAA National Centers for Environmental Information; ETOPO 2022 15 Arc-Second Global Relief Model; NOAA/NCEI; 2022. Fujii et al.; Tsunami Source of the 2011 off the Pacific Coast of Tohoku Earthquake; Earth, Planets and Space; 2011. Mori, Takahashi and The 2011 Tohoku Earthquake Tsunami Joint Survey Group; Nationwide Post Event Survey and Analysis of the 2011 Tohoku Earthquake Tsunami; Coastal Engineering Journal; 2012. Kurganov and Petrova; A Second-Order Well-Balanced Positivity Preserving Central-Upwind Scheme for the Saint-Venant System; Communications in Mathematical Sciences; 2007. Hirt and Nichols; Volume of Fluid (VOF) Method for the Dynamics of Free Boundaries; Journal of Computational Physics; 1981. Menter; Two-Equation Eddy-Viscosity Turbulence Models for Engineering Applications; AIAA Journal; 1994. Fujima, Masamura and Goto; Development of the 2D/3D Hybrid Model for Tsunami Numerical Simulation; Coastal Engineering Journal; 2002. Takase et al.; 2D–3D Hybrid Stabilized Finite Element Method for Tsunami Runup Simulations; Computational Mechanics; 2016. Watanabe et al.; Validation of Tsunami Numerical Simulation Models for an Idealized Coastal Industrial Site; Coastal Engineering Journal; 2022.